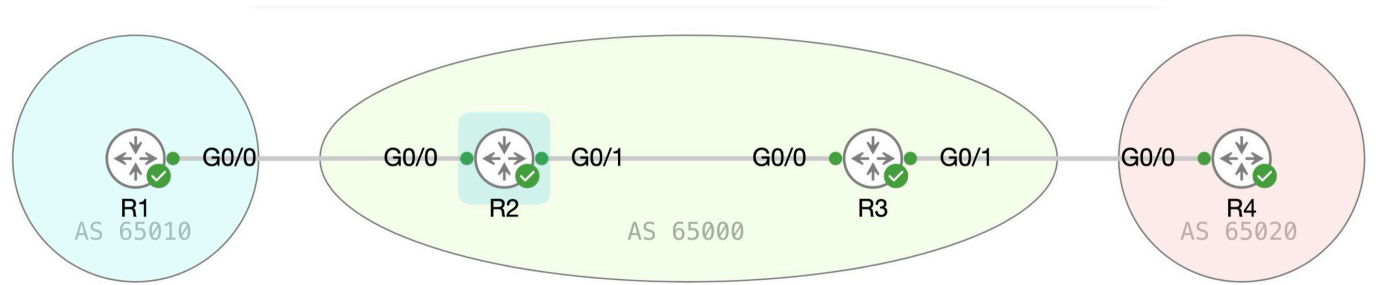


Network Topology



IP Addressing Plan

R1 to R2 - 10.0.12.0/30

R2 to R3 - 10.0.23.0/30

R3 to R4 - 10.0.34.0/30

Loopback0 - R1 - 1.1.1.1, R2 - 2.2.2.2, R3 - 3.3.3.3, R4 - 4.4.4.4

Loopback1 - sitenetwork /24 prefix R1 - 172.16.1.1/24 and R4 - 172.16.4.1/24

1 - 65010, 4- 65020, 2&3 - 65000

Router Name	Autonomous System #	Type
R1	65010	Site A Router
R2 and R3	65000	ISP Provider Routers (P Router)
R4	65020	Site B Router

Configure Interfaces

- On R1,

```
en
conf t
hostname R1

int gi0/0
ip add 10.0.12.1 255.255.255.252
no sh
exit

int loopback0
ip add 1.1.1.1 255.255.255.255
exit

int loopback1
ip add 172.16.1.1 255.255.255.0
exit
```

- On R2,

```
en
conf t
hostname R2

int gi0/0
ip add 10.0.12.2 255.255.255.252
no sh
exit

int gi0/1
ip add 10.0.23.1 255.255.255.252
no sh
exit

int loopback0
ip address 2.2.2.2 255.255.255.255
exit
```

- On R3,

```
en
conf t
hostname R3

int gi0/0
ip add 10.0.23.2 255.255.255.252
no sh
exit

int gi0/1
ip add 10.0.34.1 255.255.255.252
no sh
exit

int loopback0
ip add 3.3.3.3 255.255.255.255
exit
```

- On R4,

```
en
conf t
hostname R4

int gi0/0
ip add 10.0.34.2 255.255.255.252
no sh
exit

int loopback0
ip add 4.4.4.4 255.255.255.255
exit

int loopback1
ip add 172.16.4.1 255.255.255.0
exit
```

Protocol Configuration - OSPF, BGP and MPLS

BGP and OSPF Configuration

- On R1,

```
router bgp 65010
  neighbor 10.0.12.2 remote-as 65000
  network 172.16.1.0 mask 255.255.255.0
exit

router ospf 1
  network 0.0.0.0 255.255.255.255 area 0
exit
```

- On R2,

```
router bgp 65000
  neighbor 10.0.12.1 remote-as 65010
  neighbor 10.0.23.2 remote-as 65000
exit

router ospf 1
  network 0.0.0.0 255.255.255.255 area 0
exit
```

- On R3,

```
router bgp 65000
  neighbor 10.0.23.1 remote-as 65000
  neighbor 10.0.34.2 remote-as 65020
exit

router ospf 1
  network 0.0.0.0 255.255.255.255 area 0
exit
```

- On R4

```
router bgp 65020
  neighbor 10.0.34.1 remote-as 65000
  network 172.16.4.0 mask 255.255.255.0
exit

router ospf 1
  network 0.0.0.0 255.255.255.255 area 0
exit
```

Enable MPLS on interfaces

- On R1,

```
int gi0/0
  mpls ip
exit
```

- On R2,

```
int gi0/0
  mpls ip
exit

int gi0/1
  mpls ip
exit
```

- On R3,

```
int gi0/0
  mpls ip
exit

int gi0/1
  mpls ip
exit
```

- On R4,

```
int gi0/0
  mpls ip
exit
```